

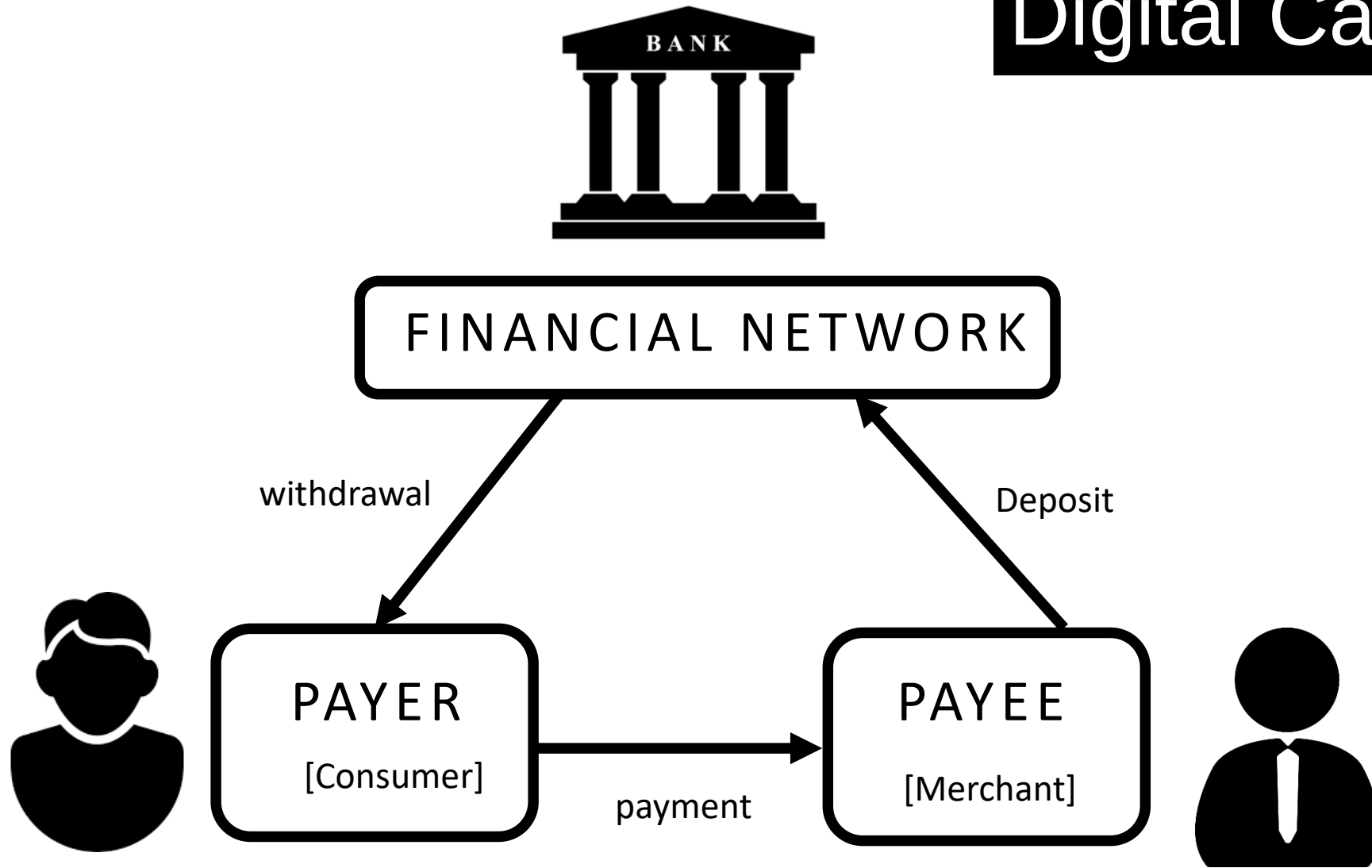
TUTORIAL – 10

Q1. Illustrate the digital cash model.

Q2. What is double spending?

How does an online payment system overcome double spending?

Digital Cash Model



Double Spending

same digital cash
used over again



\$ 50.00



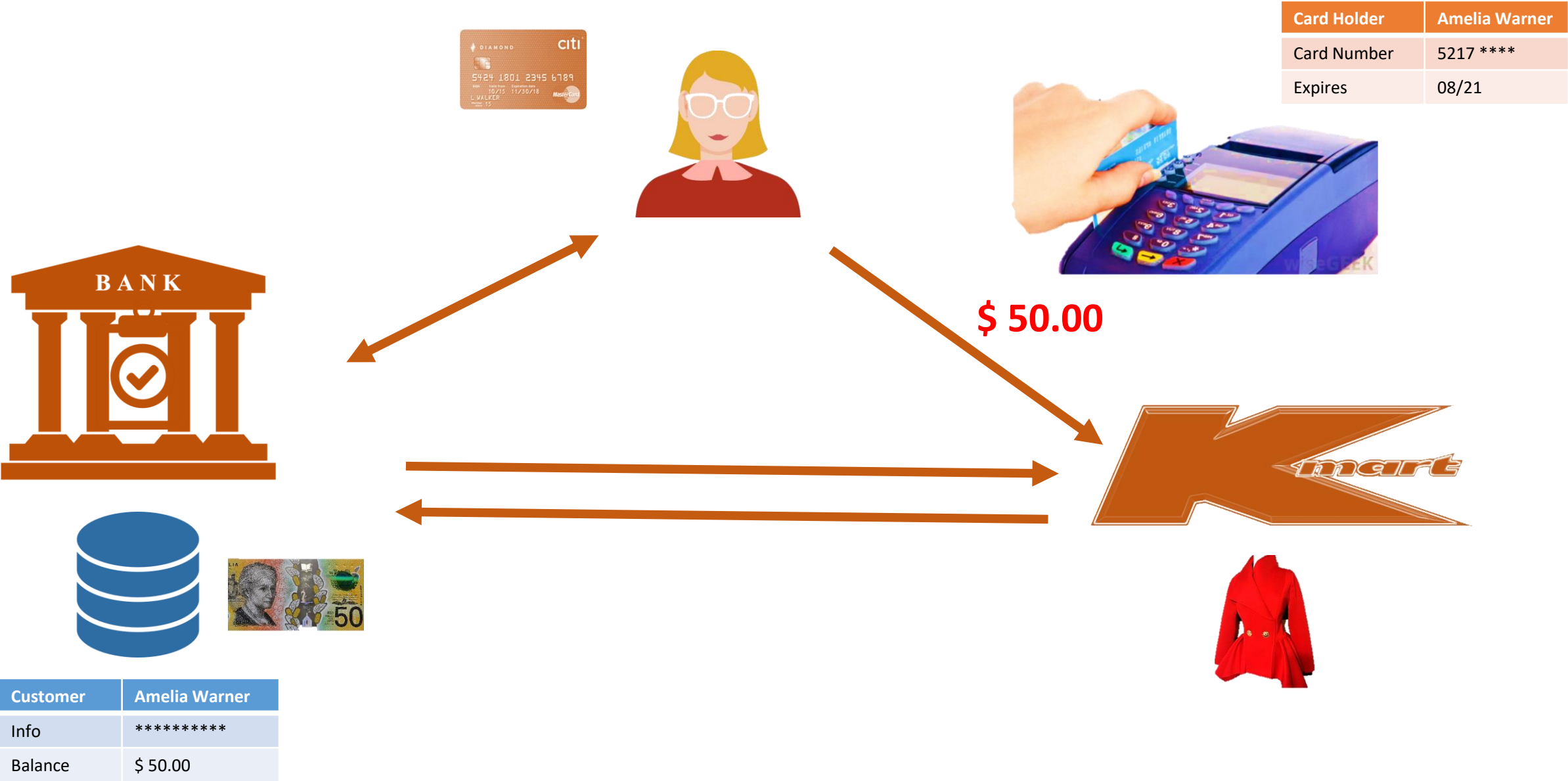
\$ 50.00



Insufficient Fund



How to Prevent Double Spending?



How to Prevent Double Spending?

- 1 Bank Maintains a database of Cash Spent
- 2 Merchant contacts Bank when Consumer pays cash
- 3 Bank checks the digital cash in the database.
- 4 Bank replies Merchant - spent or spendable

Blind Signature

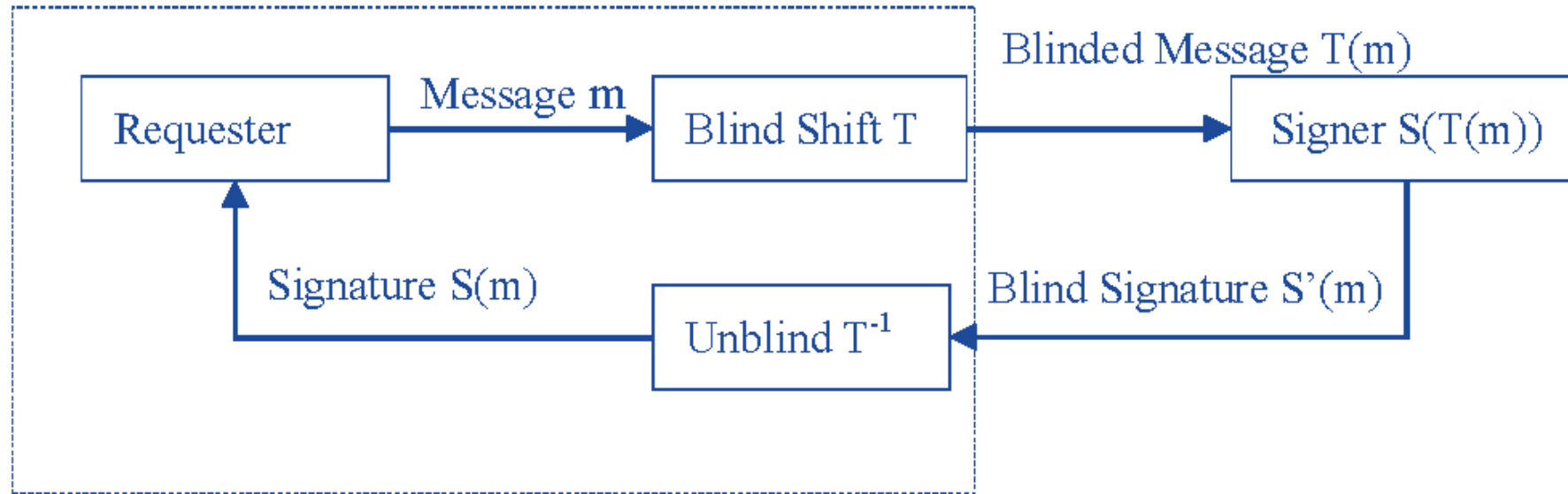
Privacy Issue of the Digital cash and Voting System



Blind Signature vs General Signature



(a) A general signature scheme



(b) The blind signature scheme

3 Steps : Blind, Sign and Unblind

Blind Signature with RSA

Blind and Unblind

Public Key (e), Private Key (d), and Mod n



Alice (Requester)

→ Sends M after binding to sign

Received signed M and unblind ←



Yuyuan (Signer)

3: Unblinds:

$$S = T^d / k = M^d k \text{ mod } n / k \\ = M^d \text{ mod } n$$

2. Signs: T using d.

$$T^d = (Mk^e)^d \text{ mod } n = M^d k \text{ mod } n$$

1. Blind: picks k ($1 < k < n$) and compute $T = Mk^e \text{ mod } n$

[Click here for RSA Basics Help](#)